



BUSINESS BLUEPRINT AOP AUTOMATION FOR NMD CRACKER PLANT

Honeywell

Site Name	Reliance Industries Ltd.
Track Name	RIL AOP Automation for NMD Cracker Plant
Rev. No.	Rev 0
Rev. Date	22-09-2025
Honeywell Doc No.	
RIL Doc No.	

Business Blueprint

AOP Automation for NMD Cracker Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

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Revision History

Rev.	Date	Originated By	Description of status and changes
0.0	22-Sep-2025	HON	Issued for Approval

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1. Introduction

The purpose of this document is to capture the detailed specifications for the Digitalized Annual Operating Plan (AOP) for NMD Cracker plant for NMD site.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various Cracker plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple plants across various sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present, this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business, as the next revised budget is expected by the end of January. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, seasonal impact on production, etc.) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for Cracker technology.

1.2 Reference Documents

Document ID	Revision	Title	Document
NMD Cracker DCT	-	Digital AOP NMD Cracker DCT	 NMD Cracker_DCT_Rev01 -
NMD Cracker Best Achieved Min CC Norms	-	NMD Cracker Best Achieved Min CC Norms	 NMD Carcker MIIS Data_Best achived_5.x
NMD Cracker Shutdown Norms	-	NMD Cracker Shutdown Norms	 Shutdown Norms Basis.xlsx

Table 1 Reference Documents

2. AOP Design Basis

The table below captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release, post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release, post review and discussion

Table 2 AOP Functional Scope

Function-wise, the detailed design basis is captured in the sections below.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Consumption AOP

Steps to generate the Production and Consumption AOP are captured in the below flow diagram.

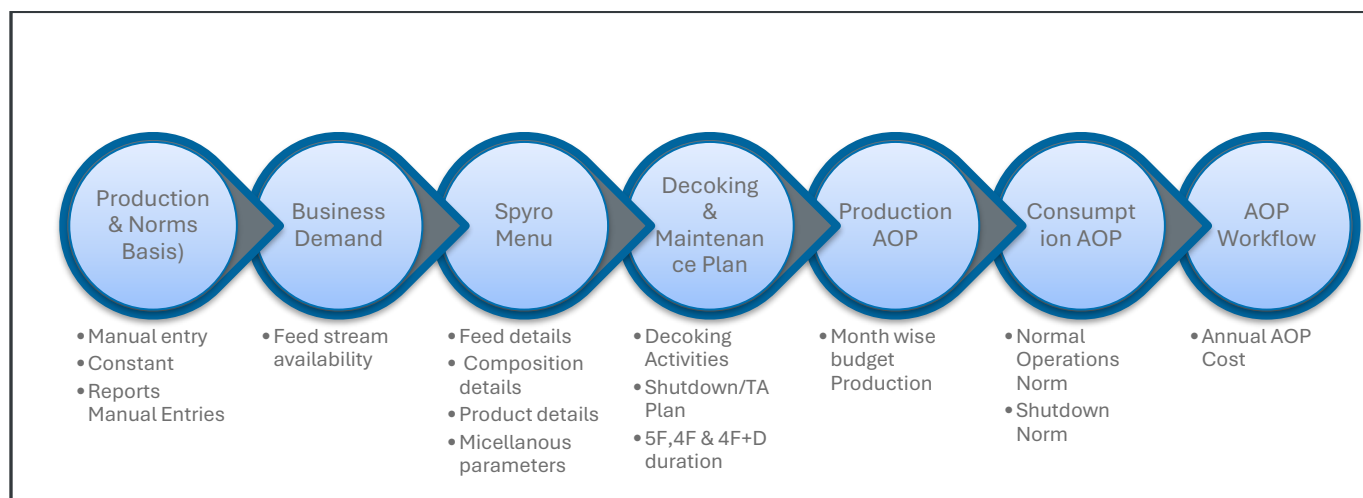


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production/Production AOP

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Production and Norms Basis

The Production and Norms Basis screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

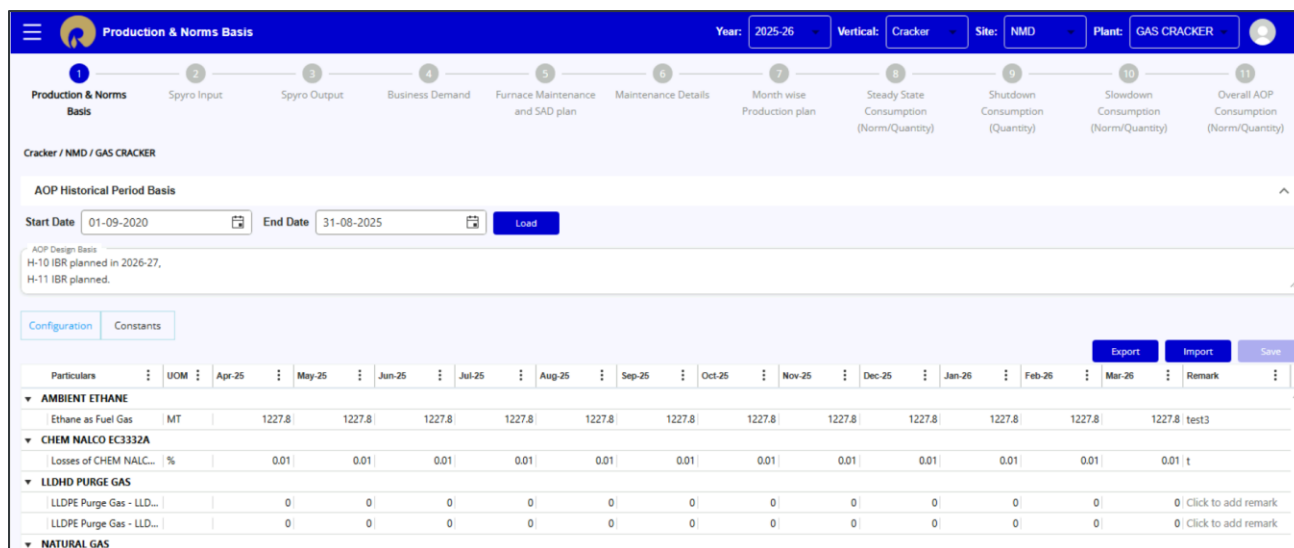
Plant-specific data, including Handling losses (chemicals), fuel calorific value, and Consumption, are configured under the Production and Norms Basis screen to support Production and Consumption Norms calculations and reports generation.

By default, the values entered for the current year will be automatically carried forward to the next year.

In Production and Norms Basis screen, below additional features and tabs are provided.

- AOP historical period basis added where Start and End date is provided to select historical period to calculate Norms data for selected period by default 05 years data will be loaded. AOP Design basis will reflect the selected period as AOP basis, users can add additional basis points for summary.
- Constant tab is provided to enter the constant value which will not change month wise as per below snapshot.
- Option to import and export manually entered from configuration screen is provided.

The screenshot below highlights the details captured in the digitized AOP platform.



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
AMBIENT ETHANE														
Ethane as Fuel Gas	MT	1227.8	1227.8	1227.8	1227.8	1227.8	1227.8	1227.8	1227.8	1227.8	1227.8	1227.8	1227.8	test3
CHEM NALCO EC3332A														
Losses of CHEM NALC...	%	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01 t
LLDHD PURGE GAS														
LLDPE Purge Gas - LLD...		0	0	0	0	0	0	0	0	0	0	0	0	Click to add remark
LLDPE Purge Gas - LLD...		0	0	0	0	0	0	0	0	0	0	0	0	Click to add remark
NATURAL GAS														

Figure 2 Digitized AOP screen for Production & Norms Basis

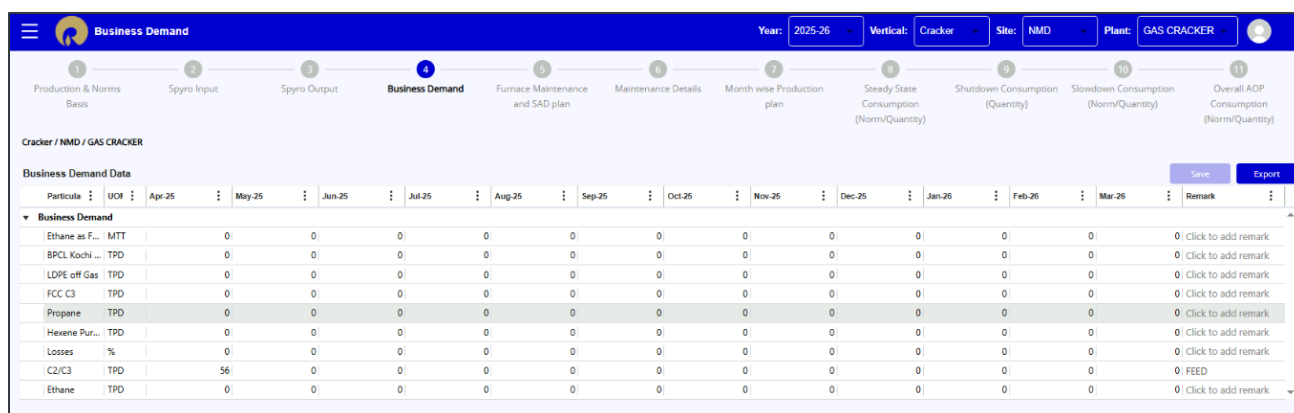
2.1.1.2 Business Demand

Business Demand will have the following items (Feed availability) which will be given / entered by Business group:

1. C2C3 Feed
2. Ethane
3. Propane
4. FCC – C3
5. BPCL Kochi Propylene
6. Ethane as Fuel Gas

Above feeds will be entered month wise. Values entered by business group in the Business Demand (Feed availability) section should be linked to Spyro Input screen

AOP digitization portal as seen in the screenshot below.



Particula	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Business Demand														
Ethane as F... MTT		0	0	0	0	0	0	0	0	0	0	0	0	0 click to add remark
BPCL Kochi... TPD		0	0	0	0	0	0	0	0	0	0	0	0	0 click to add remark
LDPE off Gas TPD		0	0	0	0	0	0	0	0	0	0	0	0	0 click to add remark
FCC C3 TPD		0	0	0	0	0	0	0	0	0	0	0	0	0 click to add remark
Propane TPD		0	0	0	0	0	0	0	0	0	0	0	0	0 click to add remark
Hexene Pur... TPD		0	0	0	0	0	0	0	0	0	0	0	0	0 click to add remark
Losses %		0	0	0	0	0	0	0	0	0	0	0	0	0 click to add remark
C2/C3 TPD		56	0	0	0	0	0	0	0	0	0	0	0	0 FEED
Ethane TPD		0	0	0	0	0	0	0	0	0	0	0	0	0 click to add remark

Figure 3 Digitized AOP screen for Business Demand

2.1.1.3 Spyro menu

Spyro menu will consist of Spyro input and output screens.

Following tabs are configured under Spyro input screen.

1. Feed
2. Composition
3. Recovery Model
4. Optimizing variables
5. Major Furnace Constraint

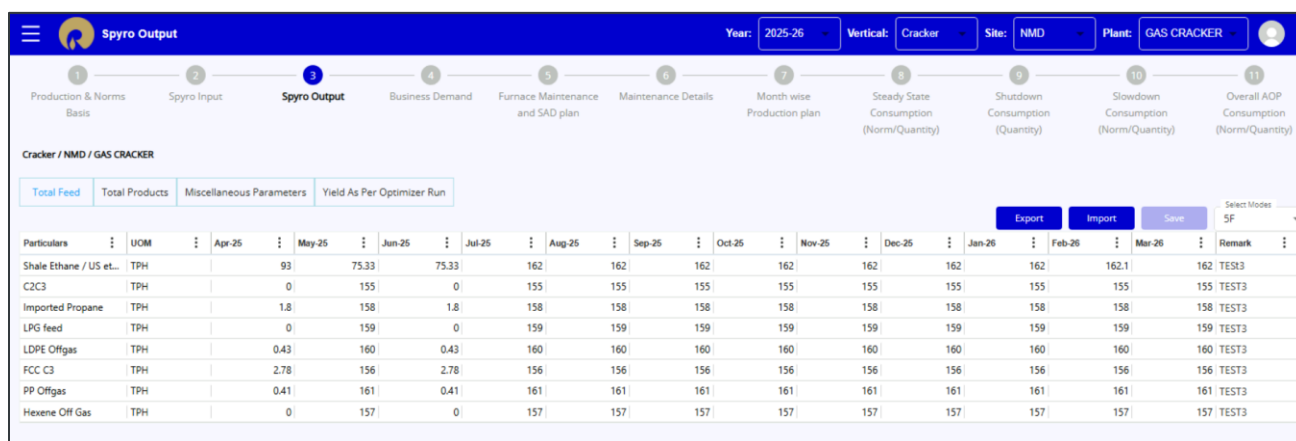


Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Ethane	TPD	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	15.3	3 TEST3
Propane	TPD	45	17.3	17.3	17.3	17.3	17.3	17.3	17.3	17.3	17.3	17.3	17.3	2 TEST5
C2/C3	TPD	56	0	0	0	0	0	0	0	0	0	0	0	0 FEED
FCC C3	TPD	5	5	5	5	5	5	5	5	5	5	5	5	5 TEST
BPCL Kochi Propylene	TPD	9	9	9	9	9	9	9	9	9	9	9	9	9 TEST
LDPE Offgas to NCP	TPD	1.76	1.5	1.69	1.49	1.76	0.98	0	0	0	0	0	0	0 TEST
PP OFF GAS	TPD	1.49	1.58	1.56	1.62	1.68	1.09	0	0	0	0	0	0	0 TEST
Hexene Purge Gas	TPD	7	7	7	7	7	7	7	7	7	7	7	7	7 TEST
Loss coefficient	%	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5 test2
Losses	%	0.58	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.03 Click to add re...

Figure 4 Digitized AOP screen for Spyro Input

Following tabs are configured under Spyro output screen.

1. Total Feed
2. Total Products
3. Miscellaneous parameters
4. Yield as per Optimizer run



Particulars	UOM	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Remark
Shale Ethane / US et...	TPH	93	75.33	75.33	162	162	162	162	162	162	162	162.1	162	162 TEST3
C2C3	TPH	0	155	0	155	155	155	155	155	155	155	155	155	155 TEST3
Imported Propane	TPH	1.8	158	1.8	158	158	158	158	158	158	158	158	158	158 TEST3
LPG feed	TPH	0	159	0	159	159	159	159	159	159	159	159	159	159 TEST3
LDPE Offgas	TPH	0.43	160	0.43	160	160	160	160	160	160	160	160	160	160 TEST3
FCC C3	TPH	2.78	156	2.78	156	156	156	156	156	156	156	156	156	156 TEST3
PP Offgas	TPH	0.41	161	0.41	161	161	161	161	161	161	161	161	161	161 TEST3
Hexene Off Gas	TPH	0	157	0	157	157	157	157	157	157	157	157	157	157 TEST3

Figure 5 Digitized AOP screen for Spyro Output

2.1.1.4 Furnace Maintenance and SAD Plan

Furnace maintenance and SAD plan consists of entry for furnace maintenance start date and maintenance end date. Remarks column for each furnace is added for mentioning the purpose of furnace shutdown (Maintenance can be for IBR, Coil replacement or Plant Shutdown).

For each furnace below two entry is provided in "SD/TA activities" table

- a) Proposed run length pre-coil replacement

b) Proposed run length post-coil replacement

Furnace run length table will have side by side columns for Actual run length and Proposed AOP. Actual column will populate data from IP21 tags and it will not be editable. Proposed AOP run length will be calculated based on run length entries made in SD/TA activities table.

The below screenshot depicts the Furnace maintenance and SAD plan captured in the AOP digitized platform.

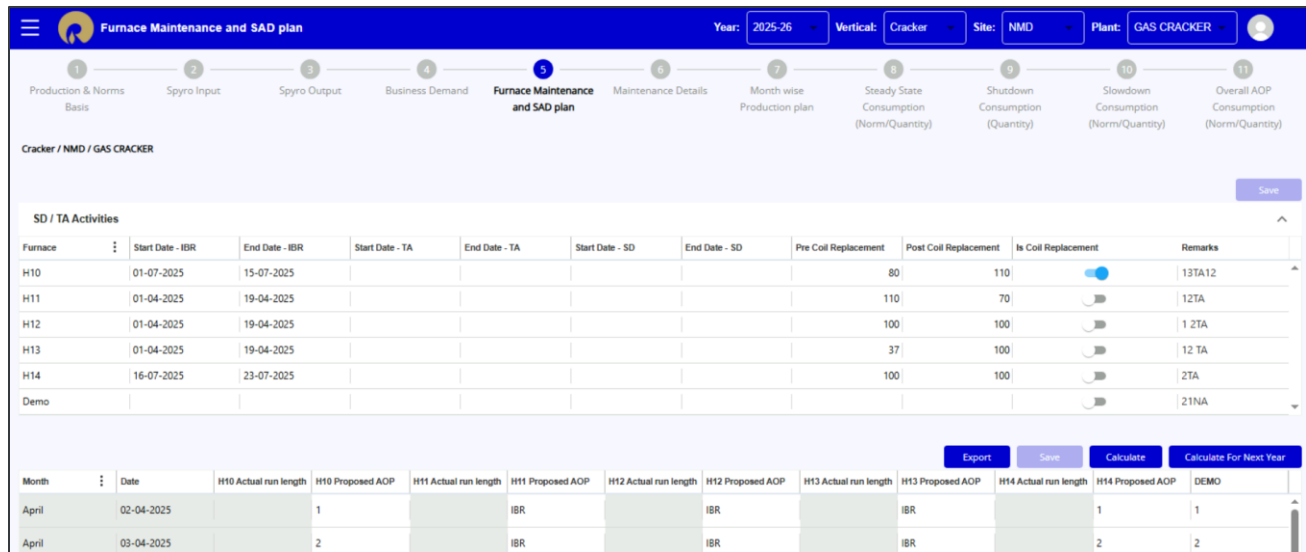


Figure 6 Digitized AOP screen for Furnace Maintenance and SAD plan

2.1.1.5 Maintenance Details

Based on the Decoke plan following parameters will reflect in Tabular form for each month in Maintenance details.

- 1) 4F days
- 2) 4F+D days
- 3) 5F days
- 4) IBR days
- 5) Shutdown days
- 6) SAD days
- 7) BBU days
- 8) HSS days
- 9) BBD days
- 10) Demo HSS days
- 11) Demo BBU days

12) Demo BBD days

13) Demo SAD days

The below screenshot depicts the Maintenance activities captured in the AOP digitized platform.

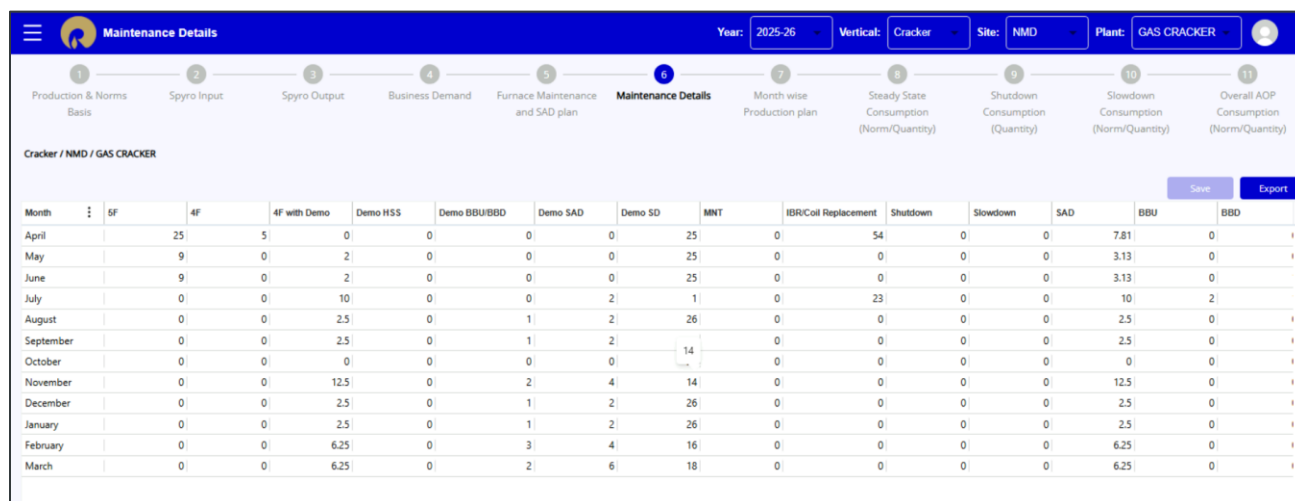


Figure 7 Digitized AOP screen for Maintenance Details

2.1.1.6 Month wise Production plan

Using the Business Demand, spyro menu, and decoke activities the month-wise production quantity is derived.

The month-wise budget production for each product is calculated in MT and KT, the formula for which is mentioned below.

$$\text{Ethylene (MT)} = \text{Ethylene based on Spyro} - \text{C2R Consumption}$$

$$\text{Proylene (MT)} = \text{Propylene based on Spyro} - \text{C3R Consumption}$$

Where,

$$\begin{aligned} \text{Ethylene based on Spyro} = & (\text{Ethylene quantity in 5F mode of operation} * \\ & \text{no of days of 5F Operation in month}) + (\text{Ethylene quantity in 4F mode of operation} * \\ & \text{no of days of 4F Operation in month}) + (\text{Ethylene quantity in 4F + D mode of operation} * \\ & \text{no of days of 4F + D Operation in month}) \end{aligned}$$

$$\begin{aligned} \text{Propylene based on Spyro} = & (\text{Propylene quantity in 5F mode of operation} * \\ & \text{no of days of 5F Operation in month}) + (\text{Propylene quantity in 4F mode of operation} * \\ & \text{no of days of 4F Operation in month}) + (\text{Propylene quantity in 4F + D mode of operation} * \\ & \text{no of days of 4F + D Operation in month}) \end{aligned}$$

This information is captured in the AOP digitization portal as seen in the screenshot below.

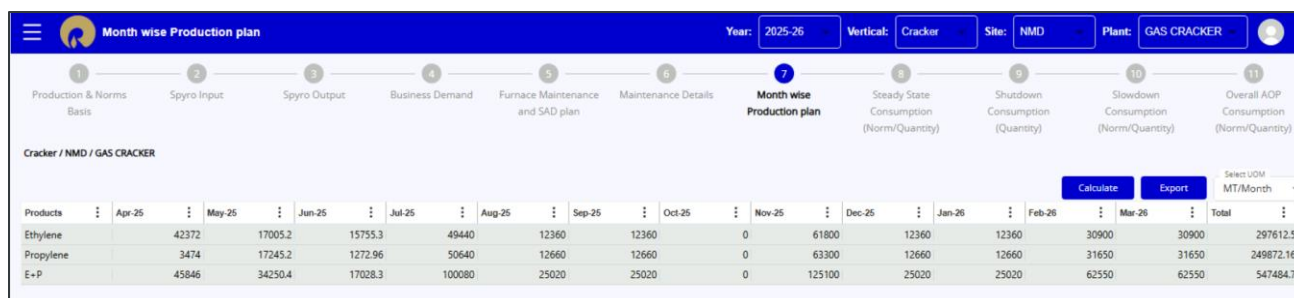


Figure 8 Digitized AOP screen for Month wise Production Plan

The '**Calculate**' button on this screen allows users to recalculate Production values when changes are made to Spyro input, Spyro output, business demand, or when furnace mode of operation days are changed in maintenance details.

2.1.2 Norms

Below are the steps captured to derive the Norms value.

2.1.2.1 Steady State Norms

For calculation purpose in the Digitized AOP platform, based on the DCT provided by the user, Norms are broadly classified as –

1. Best Achieved using minimum conversion cost
2. Based on Calculations
3. Individual Best Achieved

2.1.2.1.1. Best Achieved (Minimum Conversion Cost)

For NMD Cracker plant, last 05 years of MIIS data is considered as basis raw data to derive the utilities norms and similarly SAP data is considered to derive the Cat-chem norms.

Below is the logic formulated to derive the Historical best achieved norms using minimum conversion cost approach.

1. Day wise Consumption and Production data to be fetched from MIIS/SAP.
2. Sorting of the consumption / production data to be done based on following criteria.
 - a) Mode of Operation - 5F, 4F+D and 4F
 - b) Propane cracking - Minimum Propane (1.8 TPH), 1 Zone propane (8TPH) and 2 Zone Propane (16 TPH)
 - c) C2C3 cracking - NIL cracking, Min to Max cracking
 - d) Maximum Ethylene production
 - e) Seasonal Impact - Summer/Monsoon -Apr to Sept and Winter-Oct to Mar: 6 months each

3. Averaging and grouping of the data to be done based on the above criteria.
4. Best achieved data should be average of minimum 5 consecutive days.
5. Best achieved will be average of minimum 3 conversion consecutive data set.
6. Propane – 1 ZONE and Propane – 2 ZONE is calculated mode wise and based on the avg of minimum 03 conversion cost values, best achieved quantity will be shown for the respective time period (month) of the season.

2.1.2.1.2. Based on Calculations

For NMD Cracker Raw Material, by product, cat-chem and utilities norms are derived using calculations provided by the user in the DCT.

E.g. the Raw Material norms for 'LDPE Offgas to NCP' and 'FCC-C3' for NMD Cracker are derived using the calculations mentioned in the below table.

Plant	Category	Particular	Calculation
NMD Cracker	Raw Material	LDPE Offgas to NCP	$(x * \text{no of days of 5F Operation} + y * \text{no of days of 4F Operation} + z * \text{no of days of 4F+D Operation}) / E+P \text{ from Production AOP}$ Where, x,y & z are LDPE offgas quantity for 5F, 4F & 4F+D modes respectively.
		FCC-C3	$(x * \text{no of days of 5F Operation} + y * \text{no of days of 4F Operation} + z * \text{no of days of 4F+D Operation}) / E+P \text{ from Production AOP}$ Where, x,y & z are FCC C3 quantity for 5F, 4F & 4F+D modes respectively..

Table 3 Raw Material Norms calculations

Figure 9 Digitized AOP screen for Steady State Norms

The **'Calculate'** button on the screen will enable if the user makes any change in the dependent inputs in the Configuration screen.

The **'Save'** button on the above screen enables the user to save any norm value selected from check box from available section.

2.1.2.1.3. Individual Best Achieved

Individual best achieved norms are derived by taking monthly average norms of smallest top 3 norms. Below lower limit criteria are applied to filter the monthly consumption quantity for Cat-Chem materials.

SAP Code	Particulars	Lower Limit (kg/month)
1000003726	CHEM CYCLO HEXY	350
1000000068	CHEM DI-METHYL DI-SULPHIDE	10000
1000000086	CHEM TRISODIUM PHOSPHATE	200
1000009928	DISODIUM HYDROGEN PHOSPHATE ANHYDROUS	No Limit
1000013813	CHEM EC3269A	100
1000024816	CHEM NH3 SOLN 10%	1000
1000001022	PHENYLENEDIAMENE DORF410C;FEEDINHIBITOR	25
1000013724	CHEM EC1405A	100
1000012313	CHEM AUQAMAX (TM) EC6577A	50
1000024974	NALCO EC9078A	No limit
1000000470	CHEM MORPHOLENE	300
1000012373	CHEM ACTRENE(R) EC3268A	150
HCLV2	HYDRO CHLORIC ACID (30%) -VIRGIN	5000

SAP Code	Particulars	Lower Limit (kg/month)
1000003785	CHEM TERTIARY BUTYL CATHECOL	120
1000003820	MONO ETHANOL AMINE BASF	500
1000024994	ARSENAL EC3430A	No Limit
1000014823	CHEM EC3397A	No Limit
1000012446	CHEM NALCO EC3332A	300
1000007390	KEM WATREAT B 70M	50
CSLYE	CAUSTIC SODA LYE – GRADE 1	43000

Table 4 Cat-chem lower limit criteria

2.1.2.2 Shutdown Norms

The Shutdown norms are recorded for the months in which shutdown activities are planned, as specified in the Decoking activities screen.

Below table represents Utilities considered for TA period.

Utilities (ISBL+OSBL)		UoM
Power	ISBL+OSBL	kW
Compressed air	Plant Air+ Inst. Air+ Service Air	Nm3
Cooling water	ISBL+OSBL	m3
DM water import	ISBL+OSBL	m3
Nitrogen	ISBL+OSBL	Nm3
Service water	ISBL+OSBL	m3
Steam		
VHP	ISBL	MT
HP	ISBL	MT
MP	ISBL+OSBL	MT
LP	ISBL+OSBL	MT
Total Steam	Sum (VHP+HP+MP+LP)	MT

Table 5 List of Utilities consumed during TA Period

MIIS value for each utility as listed above during the period of TA will be taken. For each utility, sum of PAS values of each day (Start of TA till end of TA) will be done and accordingly sum value is converted to per day value which will be utilized for proposed TA dates. Budget Norms/Consumption AOP

2.1.2.3 Budget Norms/Consumption AOP

Using the values of Normal operations Norms, Shutdown Consumptions and Slowdown Norms, the final Consumption AOP value is calculated as below –

Consumption AOP for months without shutdown or slowdown = Normal Operations Norms

*Consumption AOP for months with shutdown or slowdown = Total Consumption/
Total production*

3. AOP Reports

Following report will be implemented as part of digital AOP workflow.

- Spyro Input
- Spyro Output
- Month wise Quantity , Tonnes / month
- Month wise raw data (Mode wise & Monthly Norms)
- Furnace Maintenance Plan
- Plant Contribution (T-21)
- Plant Contribution smry (T- 22)

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by NMD Site team

5. Appendix

5.1 A1- Furnace Mode of Operation Logic

Following logic is implemented to identifying Furnace Cracking and SAD/SD mode.

Furnace	Zone	Hydrocarbon Flow Tag (kg/h)	Cracking Mode	SAD/SD Mode (Decoking / Shutdown)
H-10	Zone 1	NC-GC-1FI1612	If (NC-GC-1FI1612 && NC-GC-1FI1622 && NC-GC-1FI1632) > 500 then H-10= "Cracking" Endif	If (NC-GC-1FI1612 && NC-GC-1FI1622 && NC-GC-1FI1632) < 500 then H-10 = "SAD/SD" Endif
	Zone 2	NC-GC-1FI1622		
	Zone 3	NC-GC-1FI1632		
H-11	Zone 1	NC-GC-1FI1712	If (NC-GC-1FI1712 && NC-GC-1FI1722 && NC-GC-1FI1732) > 500 then H-11 = "Cracking" Endif	If (NC-GC-1FI1712 && NC-GC-1FI1722 && NC-GC-1FI1732) < 500 then H-11 = "SAD/SD" Endif
	Zone 2	NC-GC-1FI1722		
	Zone 3	NC-GC-1FI1732		
H-12	Zone 1	NC-GC-1FI1812	If (NC-GC-1FI1812 && NC-GC-1FI1822 && NC-GC-1FI1832) > 500 then H-12 = "Cracking" Endif	If (NC-GC-1FI1812 && NC-GC-1FI1822 && NC-GC-1FI1832) < 500 then H-12 = "SAD/SD" Endif
	Zone 2	NC-GC-1FI1822		
	Zone 3	NC-GC-1FI1832		
H-13	Zone 1	NC-GC-1FI1912	If (NC-GC-1FI1912 && NC-GC-1FI1922 && NC-GC-1FI1932) > 500 then H-13 = "Cracking" Endif	If (NC-GC-1FI1912 && NC-GC-1FI1922 && NC-GC-1FI1932) < 500 then H-13 = "SAD/SD" Endif
	Zone 2	NC-GC-1FI1922		
	Zone 3	NC-GC-1FI1932		
H-14	Zone 1	NC-GC-1FI2012	If (NC-GC-1FI2012 && NC-GC-1FI2022 && NC-GC-1FI2032) > 500 then H-14 = "Cracking" Endif	If (NC-GC-1FI2012 && NC-GC-1FI2022 && NC-GC-1FI2032) < 500 then H-14 = "SAD/SD" Endif
	Zone 2	NC-GC-1FI2022		
	Zone 3	NC-GC-1FI2032		
Demo	Pass-1	NC-GC-1FIC7103	If (NC-GC-1FIC7103 && NC-GC-1FIC7101) > 500 then Demo = "Cracking" Endif	If (NC-GC-1FIC7103 && NC-GC-1FIC7101) < 500 then Demo = "SAD/SD" Endif
	Pass-2	NC-GC-1FIC7101		

Table 6 Furnace Cracking & SAD/SD mode logic

Mode Of Operation	Criteria
5F	If all 5 furnaces are in Cracking Mode - C and DEMO is in SAD/SD mode then 5 F operation
4F	If any 4 no. of furnace are in Cracking Mode - C, remaining furnace and DEMO in SAD/SD mode then 4 F operation
4F+D	If any 4 no. of furnace are in Cracking Mode - C and Demo in cracking mode - C, remaining furnace is in SAD/SD mode then 4 F +D operation

Table 7 Furnace mode of operation logic

5.2 A2- Furnace Run Length Tags

Following tags are configured to fetch the Furnace Run Length.

Tag	Description	UoM
NC-GC-H10 RL	H-10 Furnace Run length	Minute
NC-GC-H11 RL	H-11 Furnace Run length	Minute
NC-GC-H12 RL	H-12 Furnace Run length	Minute
NC-GC-H13 RL	H-13 Furnace Run length	Minute
NC-GC-H14 RL	H-14 Furnace Run length	Minute

Table 8 Furnace Run Length tags